7. Input–Process–Output

Input: user preferences. Process: validation + prompt construction + Gemini generation. Output: formatted weekly wellness/fitness plan.

Tabulation

Output Section	Example
Schedule	Weekly activity arrangement
Activity	Selected general activities
Recovery	Rest and recovery suggestions
Reminder	General wellness guidance

Coding / Technical Example

```
# FitBuddy technical flow
user_input = {
    "experience": "beginner",
    "activity": "walking",
    "time": "30 minutes"
}

prompt = f"Create a general wellness plan for {user_input}"
# Send prompt to Gemini from the secure backend.
# Display the generated response in the UI.
```

Picture / Diagram

```
WEEKLY PLAN → ACTIVITY → RECOVERY → WELLNESS
```

8. Validation and Error Handling

Empty fields, invalid time values and unavailable API responses should be handled with understandable messages. The system should not expose secret API keys to the browser.

Tabulation

Area	Example
User	User preferences
System	Prompt + Gemini response
Output	Formatted plan
Future	History/reminders/dashboard

Coding / Technical Example

```
# FitBuddy technical flow
user_input = {
    "experience": "beginner",
    "activity": "walking",
    "time": "30 minutes"
}

prompt = f"Create a general wellness plan for {user_input}"
# Send prompt to Gemini from the secure backend.
# Display the generated response in the UI.
```

Picture / Diagram

```
SECURE BACKEND → GEMINI → DATABASE/OUTPUT
```

Project Summary

FitBuddy – AI Fitness Plan Generator using Gemini Models is an academic Generative AI project demonstrating brainstorming, requirements analysis and project design. It can be implemented as a web application using Python/Flask, HTML/CSS/JavaScript, Gemini API integration and an optional database.

Note: FitBuddy should be presented as a general wellness/educational application, not as a medical diagnosis or treatment system.